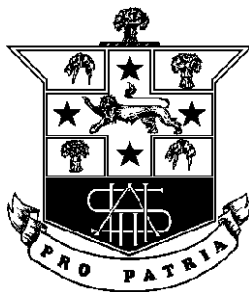


HURLSTONE AGRICULTURAL HIGH SCHOOL



MATHEMATICS – EXTENSION TWO

2006 HSC

ASSESSMENT TASK 1

Examiners ~ G Rawson, Z Pethers

GENERAL INSTRUCTIONS

- | | |
|---|---|
| <ul style="list-style-type: none">• Reading Time – 3 minutes.• Working Time – 40 MINUTES.• Attempt all questions.• All necessary working should be shown in every question.• This paper contains two (2) questions. | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved calculators may be used.• Each question is to be started on a new piece of paper.• This examination paper must NOT be removed from the examination room. |
|---|---|

STUDENT NAME: _____

TEACHER: _____

QUESTION ONE **20 marks** *Start a SEPARATE sheet*

(a) If $z = 2 - i$ and $w = 1 + 2i$, find

- | | | |
|-------|---------------|----------|
| (i) | $z + w$ | 1 |
| (ii) | $w - z$ | 1 |
| (iii) | zw | 1 |
| (iv) | $z\bar{w}$ | 1 |
| (v) | $\frac{z}{w}$ | 2 |

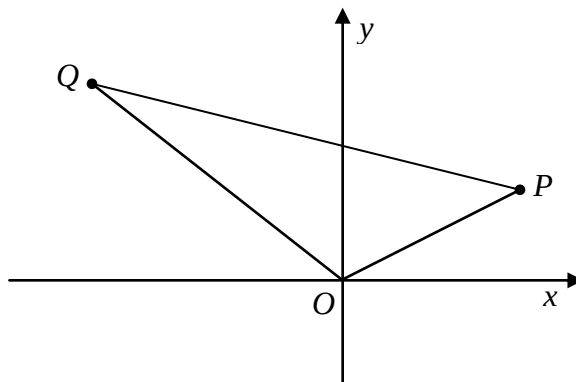
(b) Find all pairs of integers x and y that satisfy $(x + iy)^2 = 24 + 10i$ **3**

(c) Consider the equation $z^2 + az + (1 + i) = 0$.
Find the complex number a , given that i is a root of the equation **2**

(d) It is given that $2 + i$ is a root of $P(z) = z^3 + rz^2 + sz + 20$, where r and s are real numbers

- | | | |
|------|--|----------|
| (i) | State why $2 - i$ is also a root of $P(z)$ | 1 |
| (ii) | Factorise $P(z)$ over the real numbers | 2 |

(e) The diagram below show a complex plane with origin O .
The points P and Q represent arbitrary non-zero complex numbers z and w respectively.
Thus the length of PQ is $|z - w|$



- | | | |
|-------|--|----------|
| (i) | Copy the diagram, and use it to show that $ z - w \leq z + w $ | 2 |
| (ii) | Construct the point R representing $z + w$.
What type of quadrilateral is $OPRQ$? | 2 |
| (iii) | If $ z - w = z + w $, what can be said about the complex number $\frac{w}{z}$? | 2 |

QUESTION TWO **20 marks** *Start a SEPARATE sheet*

(a) Given $z = \sqrt{6} - \sqrt{2}i$, find

- | | | |
|-------|----------------------------|----------|
| (i) | $\operatorname{Re}(z^2)$ | 1 |
| (ii) | $(\operatorname{Im} z)^2$ | 1 |
| (iii) | $ z $ | 1 |
| (iv) | $\arg z$ | 2 |
| (v) | z^4 in the form $x + iy$ | 2 |

(b) Let $\alpha = 1 + \sqrt{3}i$ and $\beta = 1 + i$

- | | | |
|-------|--|----------|
| (i) | Find $\frac{\alpha}{\beta}$ in the form $x + iy$ | 2 |
| (ii) | Express α in modulus-argument form | 2 |
| (iii) | Given that β has modulus-argument form $\beta = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$,
Find the modulus-argument form of $\frac{\alpha}{\beta}$ | 2 |
| (iv) | Hence, find the exact value of $\sin \frac{\pi}{12}$ | 2 |

(c) Sketch the region in the complex plane where the inequalities

$$|z - 1 - i| < 2 \quad \text{and} \quad 0 < \arg(z - 1 - i) < \frac{\pi}{4}$$

hold simultaneously.

3

(d) The diagram below shows the locus of points z in the complex plane such that

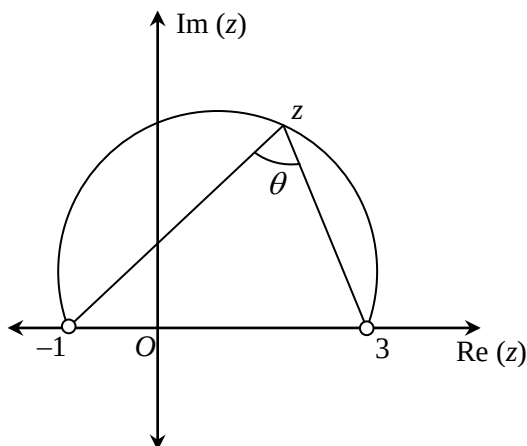
$$\arg(z - 3) - \arg(z + 1) = \frac{\pi}{3}.$$

This locus is part of a circle.

The angle between the lines from -1 to z and from 3 to z is θ , as shown.

Explain why $\theta = \frac{\pi}{3}$

2



Mathematics Ext 2 Assessment Task 1 (2006 HSC) – Q1

(a) $z = 2 - i, w = 1 + 2i$

$$(i) \quad \begin{aligned} z + w &= 2 - i + 1 + 2i \\ &= 3 + i \end{aligned}$$

$$(ii) \quad \begin{aligned} w - z &= (1 + 2i) - (2 - i) \\ &= -1 + 3i \end{aligned}$$

$$\begin{aligned} zw &= (2 - i)(1 + 2i) \\ (iii) \quad &= 2 + 4i - i - 2i^2 \\ &= 4 + 3i \end{aligned}$$

$$\begin{aligned} (iv) \quad \begin{aligned} z\bar{w} &= (2 - i)\overline{(1 + 2i)} \\ &= (2 - i)(1 - 2i) \\ &= 2 - 4i - i + 2i^2 \\ &= -5i \end{aligned} \quad \begin{aligned} (v) \quad \frac{z}{w} &= \frac{2 - i}{1 + 2i} \times \frac{1 - 2i}{1 - 2i} \\ &= \frac{-5i}{1 + 4} \\ &= -i \end{aligned} \end{aligned}$$

$$\begin{aligned} (b) \quad (x + iy)^2 &= 24 + 10i \\ x^2 - y^2 + 2xyi &= 24 + 10i \\ \therefore \begin{cases} x^2 - y^2 = 24 & \dots(1) \\ 2xy = 10 & \dots(2) \end{cases} \end{aligned}$$

Method 1:

x, y must be integers, so by inspection, $x = 5, y = 1$ or $x = -5, y = -1$

Method 2:

Substituting $y = \frac{5}{x}$ from (2) into (1):

$$\begin{aligned} x^2 - \frac{25}{x^2} - 24 &= 0 \\ x^4 - 24x^2 - 25 &= 0 \\ (x^2 - 25)(x^2 + 1) &= 0 \\ x &= \pm 5 \dots\dots (x \text{ is real}) \\ \therefore x &= 5, x = 1 \text{ or } x = -5, y = -1 \end{aligned}$$

(c) $z^2 + az + (1 + i) = 0$

Substitute $z = i$, since i is a root

$$i^2 + ai + 1 + i = 0$$

$$-1 + ai + 1 + i = 0$$

$$i(a + 1) = 0$$

$$a = -1$$

- (d) (i) Since all the coefficients of $P(z)$ are real, complex roots occur in conjugate pairs. So if $2 + i$ is a root, $2 - i$ must also be a root.

(ii) **Method 1:**

Let the other root be α , and using product of roots...

$$(2 + i)(2 - i)\alpha = -\frac{20}{1}$$

$$(4 - i^2)\alpha = -20$$

$$5\alpha = -20$$

$$\alpha = -4$$

$$\therefore P(z) = [z - (-4)][z - (2 + i)][z - (2 - i)]$$

$$= (z + 4)(z^2 - 4z + 5)$$

Method 2:

Combining the known factors:

$$(z - 2 - i)(z - 2 + i) = (z - 2)^2 + 1 = z^2 - 4z + 5$$

$$\therefore P(z) = (z^2 - 4z + 5)(z + 4)$$

where the factor $z + 4$ has been found by equating the coefficient of z^3 and the constant.

- (e) (i) The sum of any two sides of a triangle is together greater than the third side
so $PQ < OQ + OP \quad \therefore |z - w| < |w| + |z| \quad \dots \textcircled{1}$

If POQ is straight,

$$\text{then } PQ = OQ + OP \quad \therefore |z - w| = |w| + |z| \quad \dots \textcircled{2}$$

From $\textcircled{1}$ and $\textcircled{2} \quad |z - w| \leq |w| + |z| \quad \textbf{NOTE:}$ Must explain the $<$ and the $=$.

- (ii) Parallelogram

- (iii) $|z - w| = |z + w|$ means diagonals are $=$, so rectangle.

$$\text{Now, } \arg \frac{w}{z} = \arg w - \arg z = \angle QOP = \frac{\pi}{2} \text{ (rectangle)}$$

$$\therefore \frac{w}{z} \text{ is imaginary}$$

Mathematics Ext 2 Assessment Task 1 (2006 HSC) – Q2

(a) $z = \sqrt{6} - \sqrt{2}i$

$$\begin{aligned} z^2 &= (\sqrt{6} - \sqrt{2}i)(\sqrt{6} - \sqrt{2}i) \\ &= 6 - 2 - 2\sqrt{12}i \\ &= 4 - 4\sqrt{3}i \\ \text{Re}(z^2) &= 4 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad |z| &= \sqrt{(\sqrt{6})^2 + (\sqrt{2})^2} \\ &= 2\sqrt{2} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \arg z &= \tan^{-1}\left(\frac{-\sqrt{2}}{\sqrt{6}}\right) \\ &= \tan^{-1}(-\sqrt{3}) \\ &= -\frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} z^4 &= (z^2)^2 = (4 - 4\sqrt{3}i)^2 \\ \text{(iv)} \quad &= 16 - 48 - 32\sqrt{3}i \\ &= -32 - 32\sqrt{3}i \end{aligned} \quad \text{or} \quad \begin{aligned} z &= 2\sqrt{2}\text{cis}\left(-\frac{\pi}{6}\right) \\ z^4 &= (2\sqrt{2})^4 \text{cis}\left(-\frac{4\pi}{6}\right) \\ &= 64\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) \end{aligned}$$

Marks: (i) correct answer – **1 mark**

(ii) correct answer – **1 mark**

(iii) correct answer – **2 marks**; correct method with incorrect values – **1 mark** only

(iv) correct answer – **2 marks**; correct method with incorrect values – **1 mark** only

$$\begin{aligned} \text{(b)} \quad \text{(i)} \quad &\frac{1 + \sqrt{3}i}{1 + i} \times \frac{1 - i}{1 - i} \\ &= \frac{1 - i + \sqrt{3}i + \sqrt{3}}{1 + 1} \\ &= \frac{1 + \sqrt{3}}{2} + i \frac{\sqrt{3} - 1}{2} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad |\alpha| &= \sqrt{1^2 + (\sqrt{3})^2} = 2 \\ \arg \alpha &= \tan^{-1} \frac{\sqrt{3}}{1} = \frac{\pi}{3} \\ \therefore \alpha &= 2 \text{cis} \frac{\pi}{3} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \frac{\alpha}{\beta} &= \frac{2 \text{cis} \frac{\pi}{3}}{\sqrt{2} \text{cis} \frac{\pi}{4}} \\ &= \frac{2}{\sqrt{2}} \text{cis} \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\ &= \sqrt{2} \text{cis} \frac{\pi}{12} \end{aligned}$$

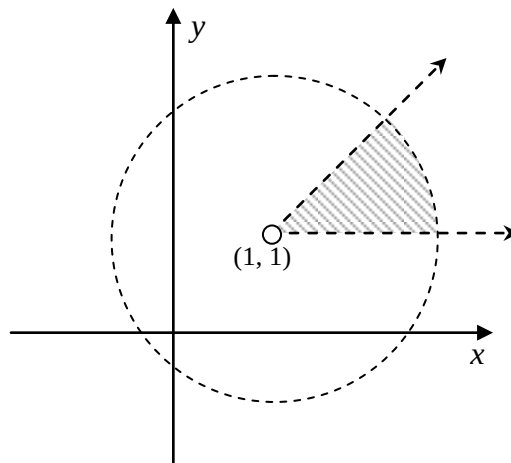
From (i) and (iii),

$$\sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right) = \frac{1 + \sqrt{3}}{2} + i \frac{\sqrt{3} - 1}{2}$$

(iv) so, $\sqrt{2} \sin \frac{\pi}{12} = \frac{\sqrt{3} - 1}{2}$

$$\sin \frac{\pi}{12} = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

(c)



(d) Let $\arg(z - 3) = \alpha$ and $\arg(z + 1) = \beta$
 $\alpha - \beta = \theta$ since the exterior angle of a triangle
is equal to the sum of the interior
opposite angles.

But we are told $\alpha - \beta = \frac{\pi}{3}$,

$$\therefore \theta = \frac{\pi}{3}$$

